

Project final report: Modeling Avalanche risk

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Deviations from Project Proposal

Several deviations from the original project proposal were introduced during the development of this project.

First, no graphical user interface (GUI) was implemented. While initially considered, the development of a user-facing interface was deemed outside the scope of a CMT project, which primarily focuses on modeling, data analysis, and algorithmic development rather than software product design.

Second, after observing that meteorological datasets provided by MeteoSwiss share a highly consistent structure and content across stations, the project scope was extended. Instead of developing a model tailored to a single location, the project was reoriented toward a **generalized framework capable of producing a customized avalanche risk prediction model for any Swiss meteorological station**, based solely on the input dataset. This design choice improves scalability and reusability without increasing model complexity.

Third, the initial proposal considered a fully discrete prediction model. This approach was ultimately abandoned in favor of a **continuous prediction model**, producing outputs in the range $[1, 5][1, 5][1, 5]$. This choice allows smoother transitions between risk levels and better captures uncertainty between discrete SLF avalanche danger classes, while remaining compatible with the official risk scale.

Finally, the set of input variables was refined during the project. MeteoSwiss provides a standardized set of meteorological variables for a large number of stations across Switzerland. To ensure generalizability and data availability, the model was designed to rely exclusively on these commonly available variables. The selected inputs include temperature, solar radiation, snow depth, a winter-day indicator, precipitation, relative humidity, and wind speed. These variables are recorded and archived on a daily basis by MeteoSwiss, enabling the construction of complete and consistent datasets for multiple regions.

Introduction to the Problem

Avalanches represent a major natural hazard in alpine regions, posing significant risks to human life, infrastructure, and economic activities. Accurate and timely assessment of avalanche danger is therefore of critical importance for public safety, mountain professionals, and decision-makers. While national services such as the Swiss Institute for Snow and Avalanche Research (SLF) provide high-quality avalanche bulletins, these assessments rely on complex expert-driven methodologies that may be difficult to reproduce or adapt locally.

This project addresses the following question:

- How can a lightweight and interpretable avalanche risk prediction model be developed using the complementary strengths of the C and MATLAB programming languages, based

on locally available meteorological data and physically meaningful relationships between weather variables and avalanche risk for Swiss mountain stations?

The motivation behind this work is twofold. First, meteorological data from MeteoSwiss are widely accessible, standardized, and consistently archived across Switzerland, making them an attractive foundation for localized risk modeling. Second, there is strong physical intuition linking meteorological conditions—such as snowfall, temperature, wind, and solar radiation—to snowpack stability and avalanche formation. Leveraging these relationships enables the construction of an empirical yet physically grounded prediction model.

Rather than aiming to replace official SLF forecasts, the objective of this project is to develop a **simple, computationally efficient, and station-specific modeling framework** that can support local analysis, exploratory studies, or educational purposes. By combining MATLAB for data processing, model calibration, and evaluation with C for efficient numerical computations, the project seeks to balance performance, transparency, and portability.

During this report, the Zermatt station specific model will be used as an example.

Approach Used

Data ingestion and preprocessing

The modeling pipeline begins with the ingestion of meteorological and avalanche risk data. Raw input datasets are first parsed and merged into a simplified table containing the target variable: SLF avalanche risk index and a selected subset of meteorological predictors. Only dates common to both datasets are retained, ensuring strict temporal alignment between inputs and outputs.

Due to missing values and quality filtering, a non-negligible number of observations must be discarded during data preparation. For transparency and model interpretability, descriptive statistics are displayed, including the number of remaining daily observations available for training and testing. This provides a clear indication of the effective dataset size on which the model is built.

Terminal visualisation: example for the Zermatt station (valais, suisse):

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➤ Station météorologique : meteo_zermatt.csv

➤ Région SLF considérée : 4222

Données restantes après préparation / nettoyage:

➤ Nombre total de jours conservés : 1974

➤ Nombre de variables météo utilisées : 13

➤ Ces données seront utilisées pour :

- l'entraînement du modèle

- le test final du modèle

=====

Feature engineering and physical consistency

To capture the complexity of avalanche processes, the model relies on engineered features derived from raw meteorological variables. These include:

- **Cumulative variables using the `movsum_c.c` function** (multi-day sums),
- **Differential variables using `diff_c.c` function,**
- **Detection of extreme events using the `movquantil_c.c` function**
- **Non-linear transformations and interaction terms.**

Feature selection and construction were guided by **physical consistency** and existing knowledge from avalanche science literature but also from direct observations during model training. I kept the variables that showed strong improvements in the model's predictions on the VALIDATION set. Numerous studies highlight the combined effects of weather variables on snowpack stability. For example:

- The interaction feature `windspeed_recent_snowfall_product` reflects the fact that fresh snow is easily transported by wind, leading to snow drift accumulation and slab formation—key triggers for avalanches.
- The squared temperature feature (`temperature2`) captures the critical role of the freezing point (0 °C), around which phase changes in the snowpack strongly influence stability.
- Multi-day aggregated variables are essential, as avalanche risk is largely determined by antecedent conditions that shape snowpack structure, cohesion, and mechanical strength rather than by same-day weather alone.

This feature engineering strategy allows the model to remain empirical while embedding physically meaningful relationships.

Dataset splitting and data leakage prevention

The prepared dataset is divided into three disjoint subsets:

- **Training set**
- **Validation set**
- **Test set**

This three-way split ensures methodological rigor. The test set is kept completely independent and is used exclusively for final performance evaluation. All model calibration, hyperparameter tuning, and design decisions are performed using the training and validation sets only, thereby preventing data leakage and optimistic bias in performance estimates.

Model formulation and training

An initial model based on **ordinary linear regression** was first implemented. However, this approach revealed a critical limitation: standard least-squares regression tends to underestimate extreme values, leading to poor performance for high-risk days.

In avalanche risk modeling, accurate detection of high-risk situations is far more important than overall average performance. A model that performs well globally but underestimates dangerous conditions would be unacceptable from a risk management perspective.

To address this issue, the model was reformulated using **Weighted Least Squares (WLS)**. In this framework, observations corresponding to high avalanche risk levels are assigned higher weights during training. The weighting scheme is defined as:

- Low-risk observations: weight = 1
- High-risk observations: weight = $1 + \alpha$

where α is a tunable hyperparameter controlling the emphasis placed on high-risk days.

Calibration of the weighting parameter (α)

Initial experiments with a fixed α value revealed strong variability in model performance across different meteorological stations. This led to the adoption of a **station-specific, data-driven calibration strategy** for α .

The optimal value of α is selected using the validation set based on the following criteria:

1. **High-risk recall constraint**

A minimum recall threshold is enforced to minimize false negatives (i.e., days where the true risk is high but predicted as low). Since missing a high-risk day represents the most severe modeling error, a minimum recall of **90%** is imposed. This threshold

was determined empirically as a compromise between safety and overall precision.

2. High-risk mean squared error (MSE_high)

An upper bound is applied to limit the average prediction error on high-risk days, ensuring numerical accuracy for the most critical situations. This threshold is motivated by the structure of the detailed SLF danger scale, which is subdivided into the levels 1, 2-, 2, 2+, 3-, 3, 3+, 4, 4+. In the continuous formulation used by the model, these subdivisions correspond to increments of approximately 0.33. I decided to set this value (one tier 0.33) as the limit for the mean error of my model in high-risk cases, so that the model remains globally close to the target. Consequently, by construction, the limit for the MSE should therefore be $0.33^2 = 0.1089$. The ideal limit would have been a semi_tier (0.165) to ensure that most of the predictions are closer to the target than to any other SLF. Unfortunately, this ideal limit could not be implemented because, during the tests I carried out, it proved to be too restrictive to maintain a consistent model for overall risk.

3. High-risk explanatory power

A constraint is applied to ensure that the coefficient of determination for high-risk observations exceeds or matches the global R^2 , guaranteeing that the model captures high-risk behavior at least as well as overall trends.

Among the α values satisfying these constraints, the final selection is made by maximizing global R^2 on the validation set, balancing overall performance and high-risk sensitivity.

Type of model

The resulting model is an **empirical, continuous-output prediction model**. It predicts avalanche risk values on a continuous scale between 1 and 5 for a particular station.

Results

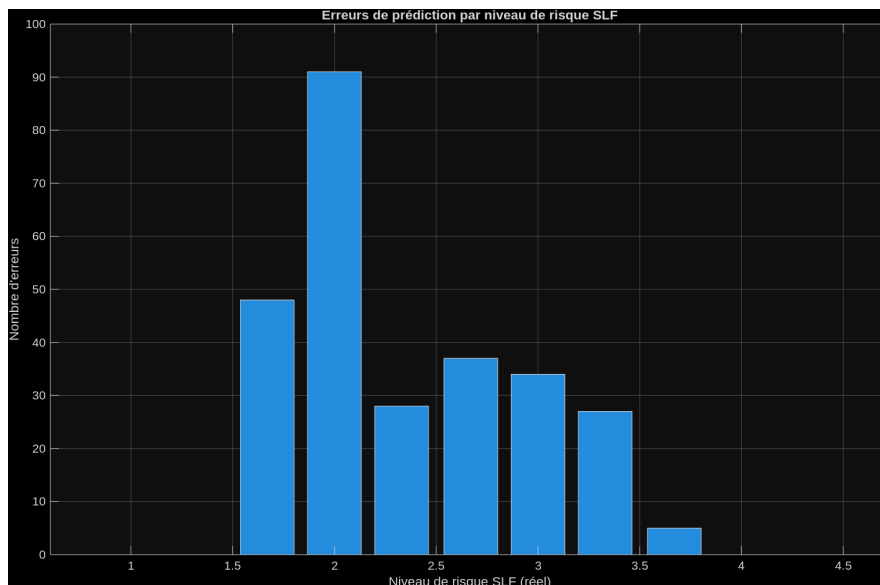
The final outcome of this project is a fully operational program capable of ingesting historical meteorological archives from a MeteoSwiss station, automatically associating the corresponding SLF avalanche risk index, and calibrating a **station-specific predictive model**. The resulting model estimates avalanche risk using a limited set of easily accessible meteorological parameters, making it both lightweight and broadly applicable.

Model outputs and visualizations

Particular attention was paid to producing **clear and minimalistic outputs** that remain informative for end users. For each station, the program exports the calibrated model coefficients, allowing users to compute avalanche risk independently once the model has been trained. All data sources used in this project are publicly available through official Swiss institutions and require no specialized access or expertise to retrieve. As a result, any user can apply the framework to estimate avalanche risk for a Swiss mountain station of their choice.

In addition to numerical outputs, the program generates several diagnostic visualizations that support a detailed interpretation of model behavior and performance:

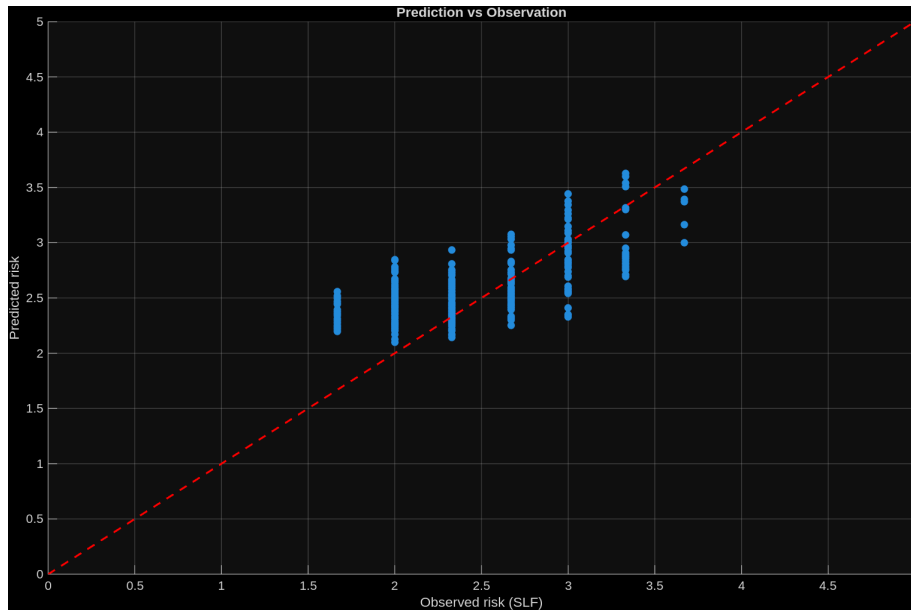
- **Prediction errors by risk level**, illustrating how model accuracy differs between low- and high-risk conditions. An error is recorded when the absolute difference between the model prediction and the SLF risk level exceeds 0.165. This threshold is motivated by the structure of the detailed SLF danger scale, which is subdivided into the levels 1, 2-, 2, 2+, 3-, 3, 3+, 4, 4+. In the continuous formulation used by the model, these subdivisions correspond to increments of approximately 0.33. Consequently, when the model's error exceeds half of this interval (0.165), the predicted value is closer to an adjacent danger tier than to the reference SLF tier.



example of prediction errors by risk level, station: Zermatt (Valais, Swiss)

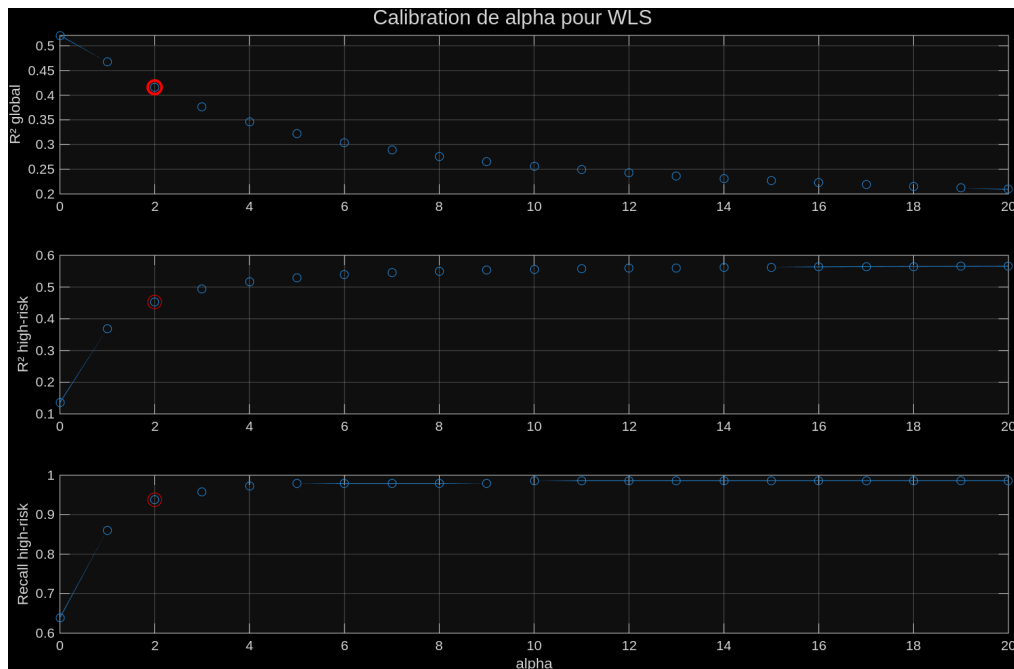
We can notice here that because of the WLS method the error number gets globally smaller as the risk increases.

- **Predicted versus observed risk plots**, illustrating deviations between model outputs and SLF reference values.



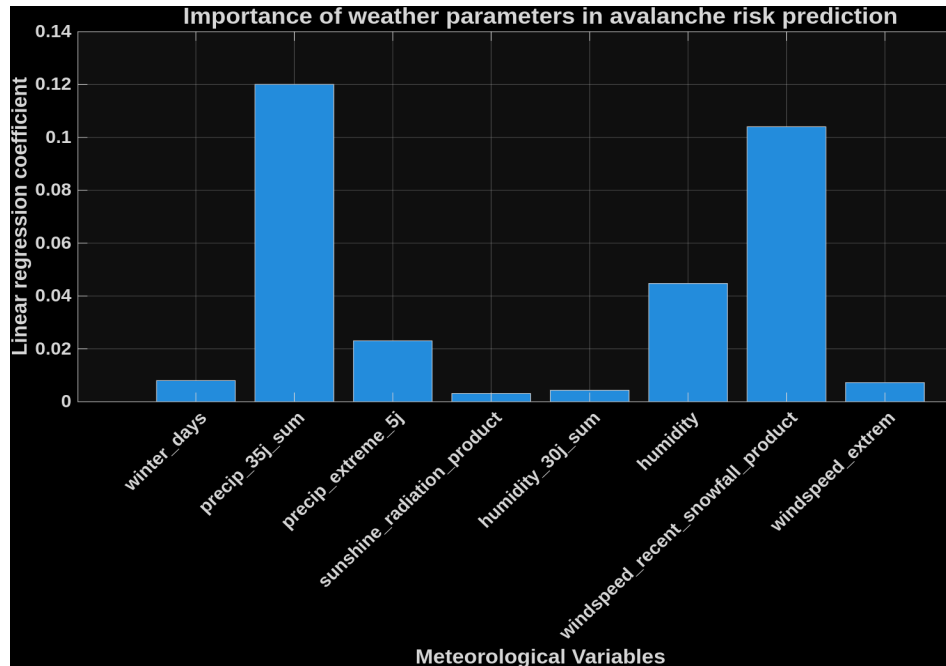
example of predicted risk versus SLF risk plot, station: Zermatt (Valais, Swiss)

- **Alpha calibration curves**, showing the impact of the weighting parameter on global and high-risk performance metrics.



example of visualisation for alpha calibration, station: Weissfluhjoch (Davos, swiss)

- **Variable importance visualizations**, providing insight into the relative influence of meteorological predictors on the estimated avalanche risk.



example of variable importance visualisation, station: Pilatus (lucerne swiss)

It is important to note that even though the overall trend in variable importance across the different station models appears scientifically sound, I observed some inconsistencies—or at least some non-obvious results—highlighted by the parameter importance plots. This issue is likely partly explained by collinearity among certain variables in my model. Indeed, if we consider, for example, the variables *snow depth* and *precipitation*, it is quite logical that their influence in the model is strongly correlated, which can sometimes lead to inconsistencies in the final coefficients. Nevertheless, I attempted to minimize this phenomenon as much as possible by retaining only the variables that are essential for the model's performance and by combining highly correlated variables, for instance with the feature *sunshine_radiation_product*, which merges two variables with very similar effects into a single parameter.

These figures enable users to identify both strengths and weaknesses of the model, fostering informed and cautious use rather than blind reliance on predictions.

Interpretation and limitations:

The interpretation of the model results must be considered in light of several important limitations related to data availability, modeling choices, and methodological assumptions.

First, the model is trained and evaluated using **SLF avalanche risk indices as the target variable**. As a direct consequence, the model **cannot provide more detailed or more accurate risk information than the SLF product itself**. In the best-case scenario, the

model is able to reproduce the same risk level as predicted by the SLF, but using a reduced and more easily accessible set of meteorological inputs. The model therefore acts as a **proxy or approximation of the SLF risk assessment**, rather than an independent avalanche forecasting system.

Then, it is important to notice that in some cases, inconsistencies emerge between theoretically expected physical relationships and the coefficients estimated by the model. Such discrepancies highlight an important limitation: avalanche risk is governed by highly nonlinear, multiscale physical processes that cannot be fully captured by a simple regression-based approach.

Consequently, while the model provides meaningful trends and robust high-risk detection, it should not be interpreted as a complete physical representation of avalanche dynamics.

Furthermore, although the meteorological data are collected at **specific weather stations located within the area of interest**, this spatial specificity is not fully exploited. SLF avalanche bulletins are issued at the **regional sector scale**, not at the individual station level. As a result, highly local meteorological variations may not be reflected in the target risk index used for training. A more locally representative target variable—such as detailed local snowpack observations or recorded avalanche events—would likely allow the station-level information to be better utilized. However, such data are considerably harder to obtain. Moreover, a model trained directly on observed avalanche occurrences, rather than on a risk index, would constitute a **fundamentally different project** with different objectives, data requirements, and evaluation criteria.

Another limitation concerns the **robustness of the model evaluation**. The current implementation relies on a single, fixed split of the dataset into training, validation, and test sets. While this approach avoids data leakage and preserves an independent test set, it introduces a degree of arbitrariness in the evaluation. Model robustness could be improved by implementing **multiple data splits or cross-validation strategies** (e.g., repeated random splits or k-fold validation). Nevertheless, the relatively large amount of available data provides a certain level of stability and reduces the risk that the reported performance is dominated by sampling variability.

Finally, the modeling approach itself introduces inherent constraints. The model is based on a **linear regression framework**, which, despite being interpretable and computationally efficient, is unlikely to fully capture the complex, non-linear nature of avalanche processes. Several avenues for improvement can be identified:

1. **Reduction of multicollinearity** among input variables, which may distort coefficient estimates and limit interpretability.
2. **More extensive exploration of non-linear transformations**, such as exponential or logarithmic functions, to better capture threshold effects and non-linear meteorological influences.
3. **Inclusion of more complex interaction terms** between variables to represent combined effects (e.g., wind–snow–temperature interactions).

4. **Use of more advanced modeling techniques** (e.g., non-linear regression, tree-based models, or probabilistic approaches), which may better represent the inherent complexity of natural avalanche phenomena.

Despite these limitations, the model provides a **solid and interpretable baseline**, demonstrating that a relatively simple, physically informed empirical approach can achieve meaningful predictive performance for avalanche risk assessment.

Conclusion and outlook

This project demonstrates that a lightweight, empirical avalanche risk prediction model can be successfully developed using publicly available meteorological data and a carefully calibrated weighted regression framework. By leveraging standardized MeteoSwiss observations and SLF avalanche risk indices, the proposed approach produces station-specific, continuous risk predictions that prioritize the accurate detection of high-risk situations—an essential requirement in avalanche hazard assessment. The use of weighted least squares and validation-driven calibration enables the model to balance global predictive performance with strong sensitivity to dangerous conditions, while remaining transparent and computationally efficient. Although limited by its linear structure and reliance on regional SLF risk indices, the model provides a robust and interpretable baseline for localized avalanche risk estimation. Overall, this work highlights the potential of physically informed empirical models as practical tools for risk-oriented applications, while clearly identifying pathways for future methodological and data-driven improvements

Bibliography

- **Mesured risks for SLF sectors:**
<https://envidat.ch/#/metadata/comparing-human-forecasts-with-model-predictions?search=avalanche&isAuthorSearch=false>
- **Assistance received** from Mr. Frank Techel, scientific collaborator at SLF in the field of avalanche risk, for understanding and interpreting the datasets.
- **MeteoSwiss meteorological measurement stations:**
<https://www.meteoswiss.admin.ch/services-and-publications/applications/ext/download-data-without-coding-skills.html#lang=en&mdt=homogenous&pgid=&sid=&col=&di=&tr=&hdr=>
- **Other sources of information used for the project:**
[The Avalanche-Hazard Index | Annals of Glaciology | Cambridge Core](#)
<https://www.research-collection.ethz.ch/entities/publication/b88321ea-803b-4dbf-bea6-b876211f4054>
[European Avalanche Warning Services](#)
[WSL Institut pour l'étude de la neige et des avalanches SLF](#)
[Bulletin d'avalanches et situation nivologique | White Risk](#)